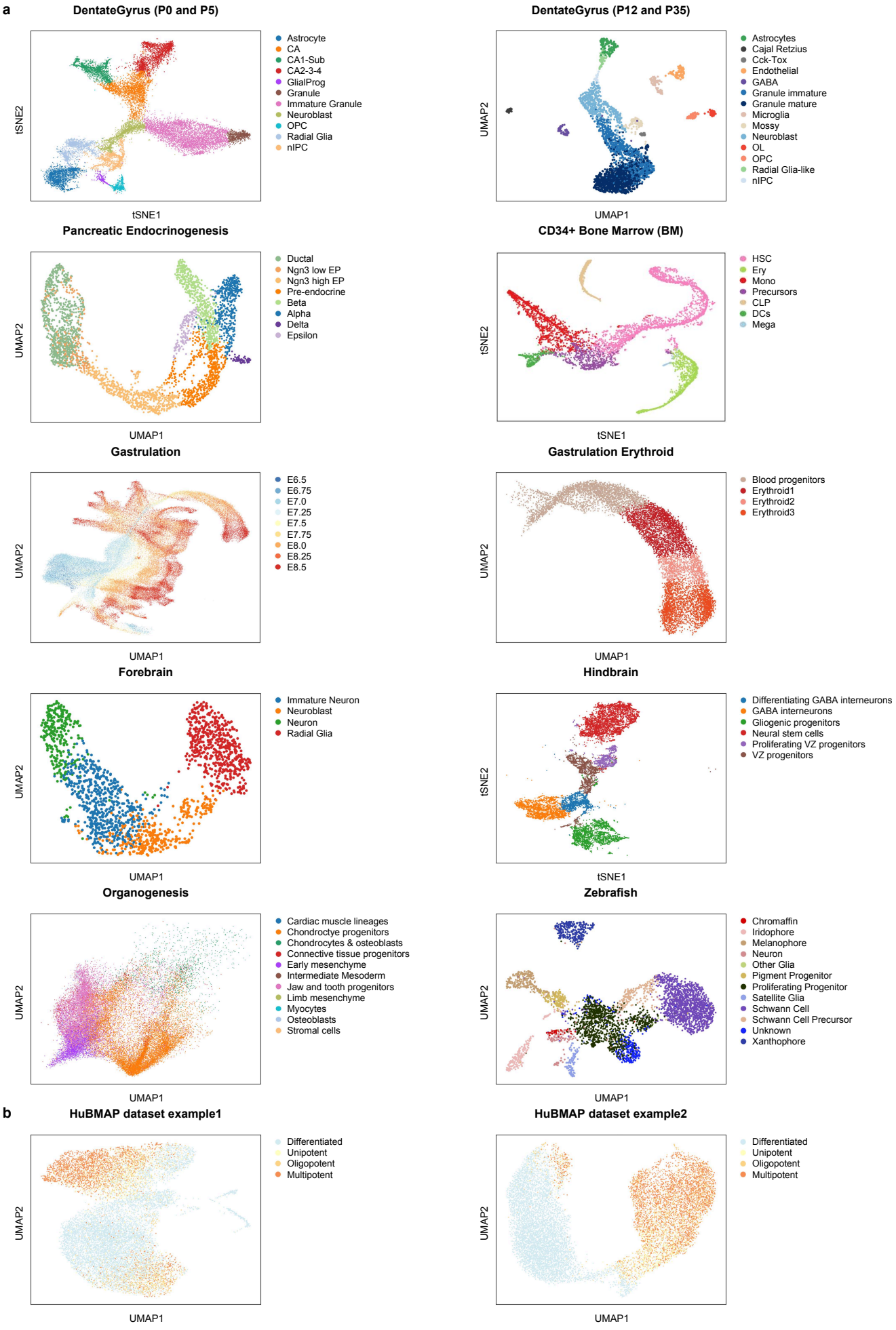

a
DentateGyrus (P0 and P5)
tSNE2
tSNE1
Astrocyte
CA
CA1-Sub
CA2-3-4
GlialProg
Granule
Immature Granule
Neuroblast
OPC
Radial Glia
nIPC
DentateGyrus (P12 and P35)
UMAP2
UMAP1
Astrocytes
Cajal Retzius
Cck-Tox
Endothelial
GABA
Granule immature
Granule mature
Microglia
Mossy
Neuroblast
OL
OPC
Radial Glia-like
nIPC
Pancreatic Endocrinogenesis
UMAP2
UMAP1
Ductal
Ngn3 low EP
Ngn3 high EP
Pre-endocrine
Beta
Alpha
Delta
Epsilon
CD34+ Bone Marrow (BM)
tSNE2
tSNE1
HSC
Ery
Mono
Precursors
CLP
DCs
Mega
Gastrulation
UMAP2
UMAP1
E6.5
E6.75
E7.0
E7.25
E7.5
E7.75
E8.0
E8.25
E8.5
Gastrulation Erythroid
UMAP2
UMAP1
Blood progenitors
Erythroid1
Erythroid2
Erythroid3
Forebrain
UMAP2
UMAP1
Immature Neuron
Neuroblast
Neuron
Radial Glia
Hindbrain
tSNE2
tSNE1
Differentiating GABA interneurons
GABA interneurons
Gliogenic progenitors
Neural stem cells
Proliferating VZ progenitors
VZ progenitors
Organogenesis
UMAP2
UMAP1
Cardiac muscle lineages
Chondroctye progenitors
Chondrocytes & osteoblasts
Connective tissue progenitors
Early mesenchyme
Intermediate Mesoderm
Jaw and tooth progenitors
Limb mesenchyme
Myocytes
Osteoblasts
Stromal cells
Zebrafish
UMAP2
UMAP1
Chromaffin
Iridophore
Melanophore
Neuron
Other Glia
Pigment Progenitor
Proliferating Progenitor
Satellite Glia
Schwann Cell
Schwann Cell Precursor
Unknown
Xanthophore
b
HuBMAP dataset example1
UMAP2
UMAP1
Differentiated
Unipotent
Oligopotent
Multipotent
HuBMAP dataset example2
UMAP2
UMAP1
Differentiated
Unipotent
Oligopotent
Multipotent

Extended Data Fig. 3 | Real-world benchmark datasets. Overview of benchmark datasets. **a,** UMAP/t-SNE illustrations of ten classical datasets, colored by cell type or time point. **b,** Two HuBMAP examples colored by CytoTRACE2 developmental potential. Quantitative method comparisons are presented in Extended Data Fig. 6. Full spatial and KPTracer datasets are shown separately in Extended Data Figs. 9 and 11.